

TOC - ch - 3

Context Free Grammar

Grammar : It is a set of rules to construct the sentences in a language.

In TOC, It is denoted as ' G '.

- It is a standard way to represent a language.
- It will tell that whether a given string is a part of language or not.

Context Free Grammar : It is a set of rules to construct the sentences in a language where each rule has only one non-terminal on the left side.
(3 Marks)
i.e. $CFG = \text{Grammar} + \text{restriction (left side must be only one symbol)}$

Terminal : final symbol $\rightarrow$ can't be replaced further

Non-terminal : Not finished / not final symbol

- It is defined as quadruple :

$$G = \{ V, T, P, S \}$$

$\downarrow$ variable $\downarrow$ Terminal $\rightarrow$ start symbol
 $\rightarrow$ production rule

ex. $S \rightarrow a S b \mid \epsilon$

$\swarrow$ start symbol & variable

$\underbrace{a S b}_{\text{production rule 1}} \quad \underbrace{\epsilon}_{\text{production rule 2}}$

$a, b \rightarrow \text{terminals}$

$\epsilon \rightarrow \text{null}$

Now, we will generate string or language based on given production rule or given grammar.

here, language = collection of strings

ex. $S \rightarrow a S b \mid \epsilon$

→ minimum string generated is ϵ

$\epsilon, a S b$

→ now, put ϵ in place of S in $a S b$

so, it will become ab

→ now, put $a S b$ in place of S

so, it will be like: $aa S b b = a^2 S b^2$

→ now, again put ϵ instead of S in $a^2 S b^2$

so, it will be like: $a^2 b^2$

→ now, put $a S b$ in place of S in $aa S b b$

so, it will become $aaa S b b b$

$$\therefore a^3 S b^3$$

now, again put ϵ instead of S in $a^3 S b^3$

$$\therefore \underline{a^3 b^3}$$

$\vdots$

$$a^4 b^4, a^5 b^5, \dots, \underline{a^n b^n}, n \geq 0$$

→ so, generated language from given

grammar is : $a^n b^n, n \geq 0$

Note : CFG is also known as Type 2 Grammar.

ex. $\left. \begin{array}{l} S \rightarrow SS \\ S \rightarrow aSb \\ S \rightarrow bSa \\ S \rightarrow \epsilon \end{array} \right\}$ multiple production rules.

→ put ϵ instead of S

$\epsilon, aSb, bSa, aSb bSa, aSb bSa aSb aSb$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $ab, ba, \underline{abba}, \underline{abbaabab}$

→ from this we can say that number of a's and number of b's are same.

Language $\Rightarrow n_a(w) = n_b(w)$ here, w is a string.

Context Free Language (CFL): It is a language generated by a CFG (Context free Grammar).
 here, language is a set of strings. (3 marks)

→ CFG : rules

CFL : output (i.e. all strings made using rules)

So, CFG generates CFL.

→ For the above given example of Grammar:

ex. $S \rightarrow aSb / \epsilon$

it can also be written like below:

$S \rightarrow aSb$

$S \rightarrow \epsilon$

→ strings generated are : $S \rightarrow \epsilon \rightarrow ""$
 $S \rightarrow aSb \rightarrow ab$
 $S \rightarrow a(aSb)b \rightarrow aabbb$
 $S \rightarrow aaabbbb$

→ So the language is : $L = \{\epsilon, ab, aabbb, aaabbbb, \dots\}$

→ this is Context Free language.

4 types of grammars :

- ① Type 0 Grammar (Unrestricted Grammar)
- ② Type 1 Grammar (Context-Sensitive Grammar)
- ③ Type 2 Grammar (Context-Free Grammar)
- ④ Type 3 Grammar (Regular Grammar)

Types of Grammars, Languages and Automata (3 Marks)

According to **Chomsky Hierarchy**, grammars, languages and automata are classified into 4 types.

Chomsky Hierarchy Table (3 Marks)

Type	Grammar	Language	Automata
Type 0	Unrestricted Grammar	Recursively Enumerable Language	Turing Machine
Type 1	Context Sensitive Grammar (CSG)	Context Sensitive Language	Linear Bounded Automata (LBA)
Type 2	Context Free Grammar (CFG)	Context Free Language (CFL)	Pushdown Automata (PDA)
Type 3	Regular Grammar	Regular Language	Finite Automata (FA)

* **Regular Grammar** : It is a type of formal grammar that generates regular languages. (3 Marks)

→ Regular Grammars can be further classified into two main types :

(1) Right Linear Grammar

- In this RLGr, the non-terminal symbol in a production rule always appears at the rightmost position.

for e.g. $A \rightarrow xB$

where, 'A' and 'B' are non-terminal symbols and 'x' is a terminal symbol.

(2) Left Linear Grammar

- In this LLGr, the non-terminal symbol in a production rule always appears at the leftmost position.

for e.g. $A \rightarrow Bx$

where, 'A' and 'B' are non-terminal symbols and 'x' is a terminal symbol.